

Setup Guide: Bumper Bowling

Time Allocation: 45 Minutes

Challenge Overview

The robot must accelerate down a lane and knock over a formation of lightweight plastic bowling pins.

Environment & Arena Setup

- **Lane Construction:** Tape a straight lane on the floor approximately 8 to 10 feet long and 2 to 3 feet wide, using painter's tape to mark the start line and boundaries.
- **Pin Formation:** Set up 6 lightweight plastic bowling pins or empty plastic bottles at the end of the lane in a standard triangle bowling formation (1 in front, 2 in the second row, 3 in the back row).
- **Starting Zone:** Designate a clear starting box at the opposite end of the lane where robots must begin stationary behind the start line.
- **Safety Boundary:** Ensure there is a buffer zone behind the pin formation to catch sliding pins and prevent them from scattering into neighboring student workspaces.